6. The value of $\log_{\sqrt{2}} 512$ is

   (1) 16 (2) 18 (3) 9 (4) 12

7. The value of $\log_3 \frac{1}{81}$ is

   (1) -2 (2) -8 (3) -4 (4) -9

8. If $\log_{\sqrt{x}} 0.25 = 4$, then the value of x is

   (1) 0.5 (2) 2.5 (3) 1.5 (4) 1.25

9. The value of $\log_a b \ \log_b c \ \log_c a$ is

   (1) 2 (2) 1 (3) 3 (4) 4

10. If 3 is the logarithm of 343, then the base is

    (1) 5 (2) 7 (3) 6 (4) 9

11. Find a so that the sum and product of the roots of the equation $2x^2 + (a-3)x + 3a - 5 = 0$ are equal is

    (1) 1 (2) 2 (3) 0 (4) 4

12. If a and b are the roots of the equation $x^2 - kx + 16 = 0$ and satisfy $a^2 + b^2 = 32$, then the value of k is

    (1) 10 (2) -8 (3) $-8, 8$ (4) 6

13. The number of solutions of $x^2 + |x - 1| = 1$ is

    (1) 1 (2) 0 (3) 2 (4) 3

14. The equation whose roots are numerically equal but opposite in sign to the roots of $3x^2 - 5x - 7 = 0$ is

    (1) $3x^2 - 5x - 7 = 0$ (2) $3x^2 + 5x - 7 = 0$ (3) $3x^2 - 5x + 7 = 0$ (4) $3x^2 + x - 7$

15. If 8 and 2 are the roots of $x^2 + ax + c = 0$ and $3, 3$ are the roots of $x^2 + dx + b = 0$, then the roots of the equation $x^2 + ax + b = 0$ are

    (1) $1, 2$ (2) $-1, 1$ (3) $9, 1$ (4) $-1, 2$

16. If a and b are the real roots of the equation $x^2 - kx + c = 0$, then the distance between the points $(a, 0)$ and $(b, 0)$ is

    (1) $\sqrt{k^2 - 4c}$ (2) $\sqrt{4k^2 - c}$ (3) $\sqrt{4c - k^2}$ (4) $\sqrt{k - 8c}$